covid | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection)

covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

covid | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

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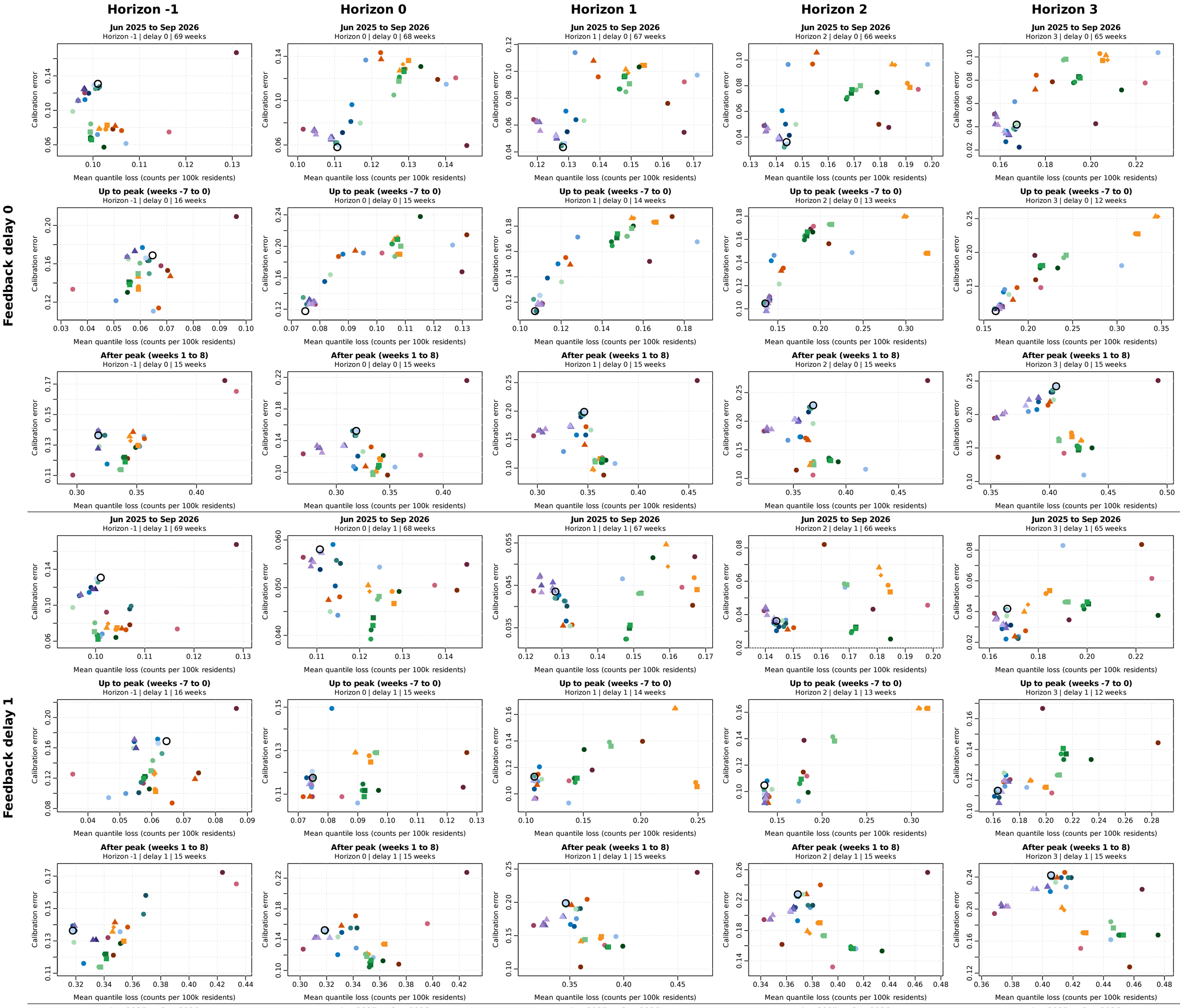
covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: soft sw
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
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Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection)

covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4) (w 0.003, leak 0.1)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, three signals (B, strength=1, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, three signals (B, strength=1, no sharing)
- Score experts: soft switch, target signal (C, delta=0, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

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Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
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With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection)

covid | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

flu | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo
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flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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Rate-grid experts
- Rate grid: equal
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slow 0.003, leak 0.1)
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Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0.1, H=4, sharing 1/8)
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- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, target signal (C, delta=0.1, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

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flu | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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- Score experts: context vectors, blended (D, delta=0, no sharing)

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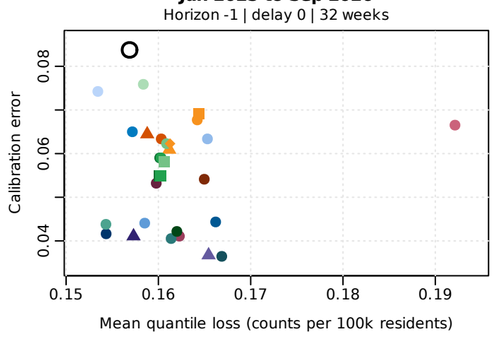
flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

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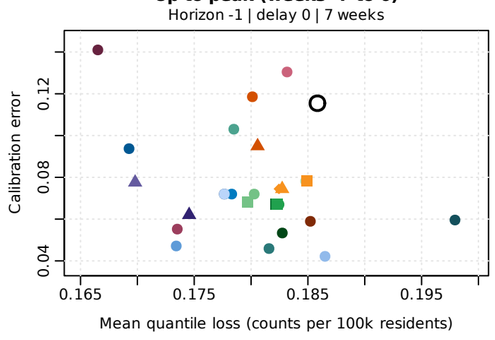
Horizon -1

Feedback delay 0

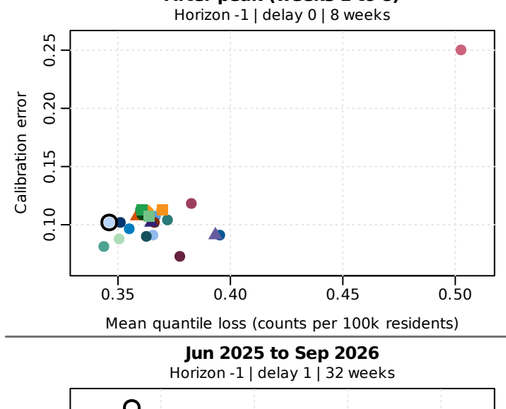
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks

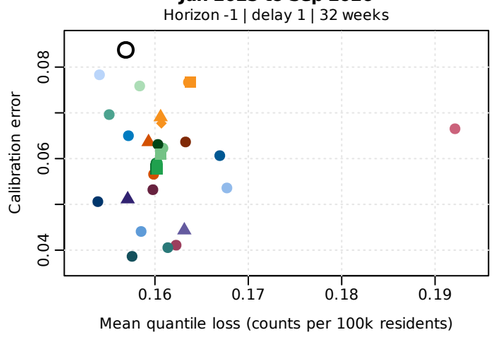


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

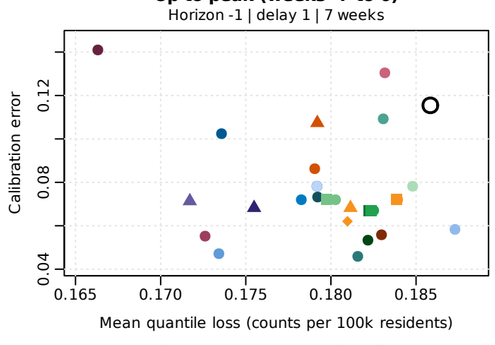


Feedback delay 1

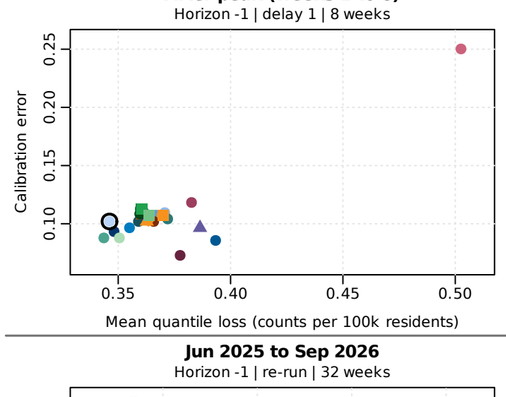
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks

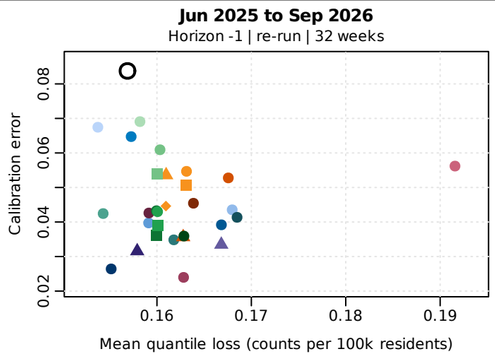


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

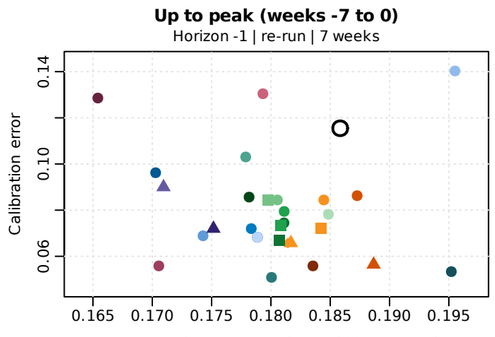


Feedback delay 0 + revisions (re-run)

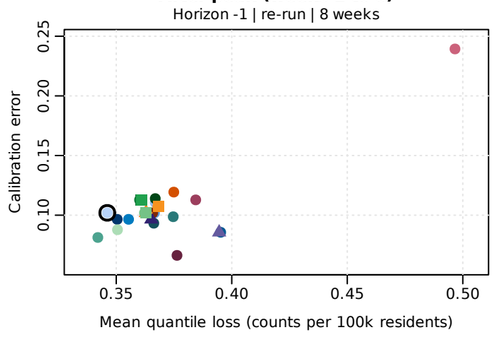
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks



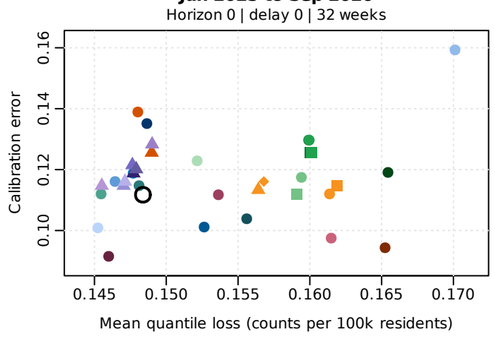
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



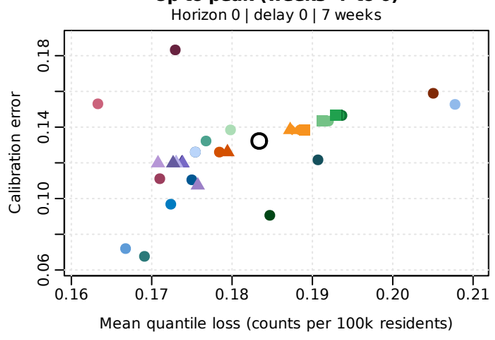
Horizon 0

Feedback delay 0

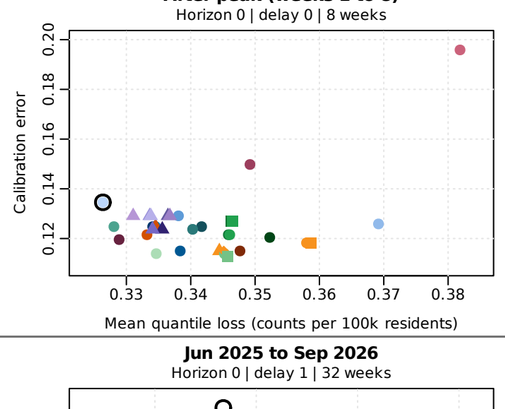
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks

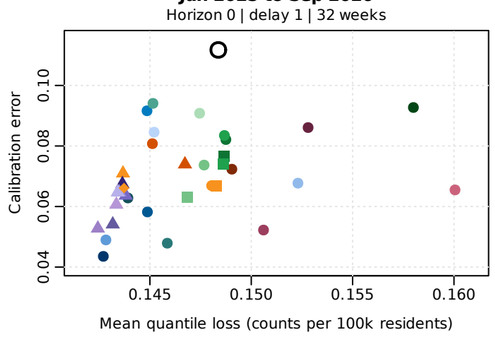


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

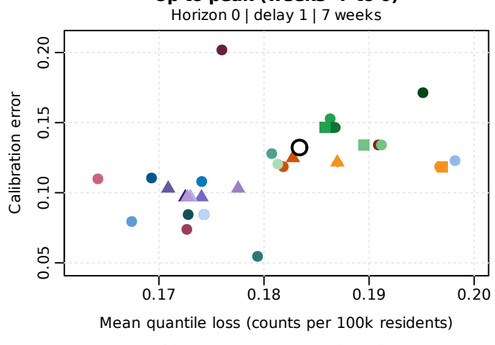


Feedback delay 1

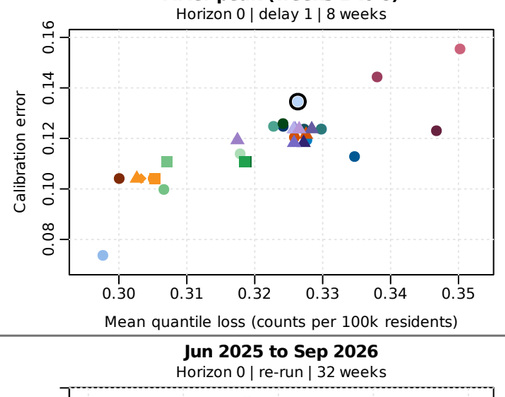
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks

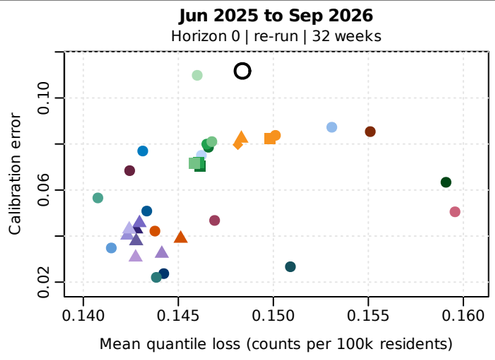


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

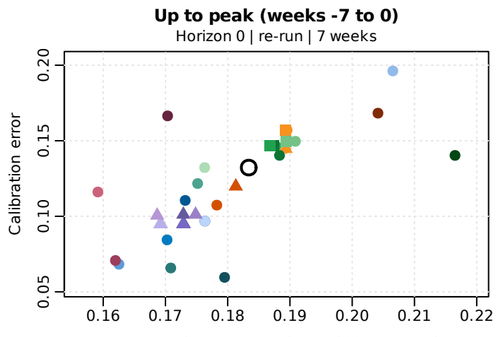


Feedback delay 0 + revisions (re-run)

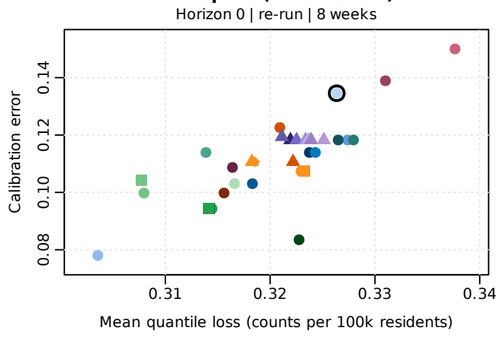
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks



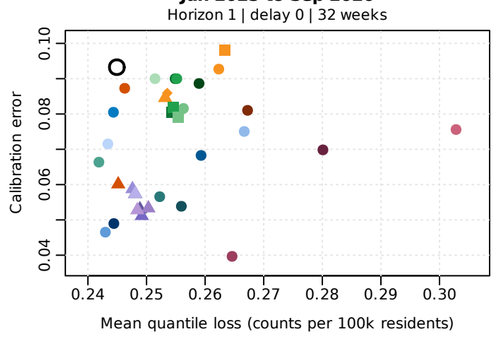
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



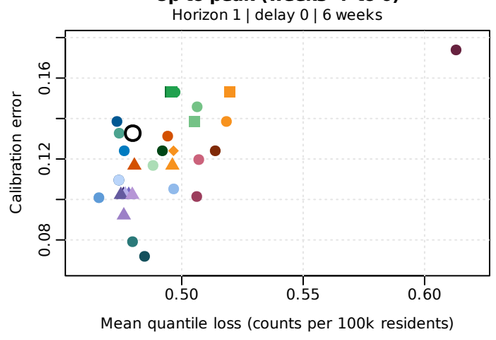
Horizon 1

Feedback delay 0

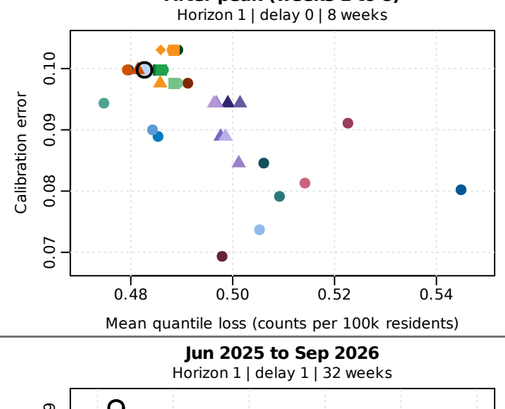
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks

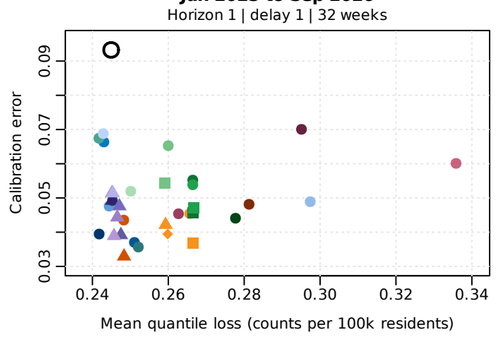


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

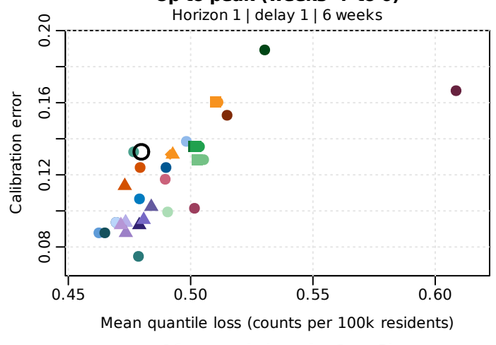


Feedback delay 1

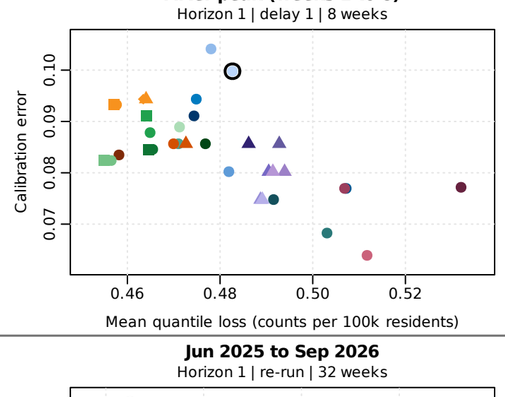
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks

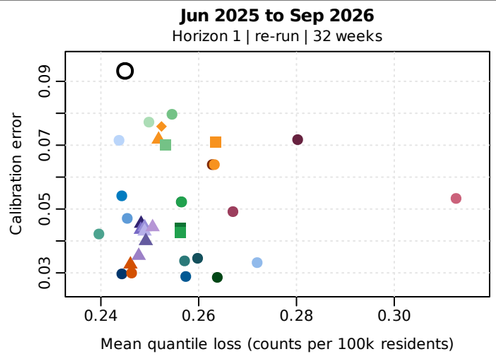


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

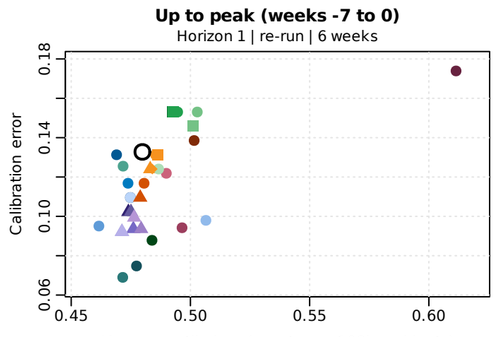


Feedback delay 0 + revisions (re-run)

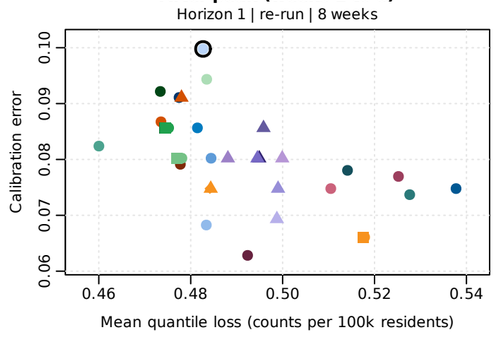
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks



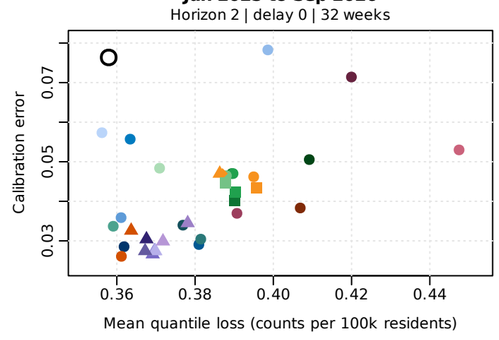
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



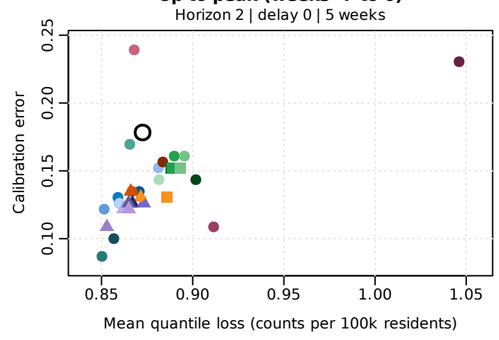
Horizon 2

Feedback delay 0

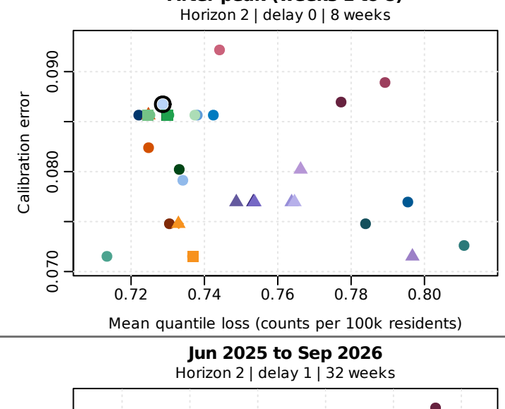
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks

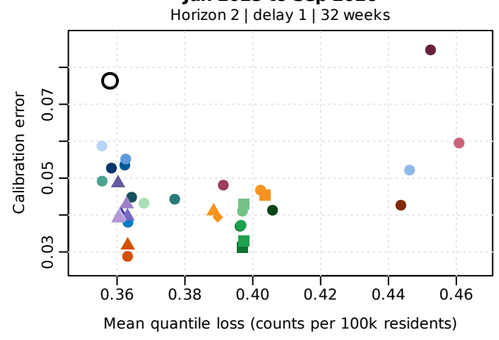


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

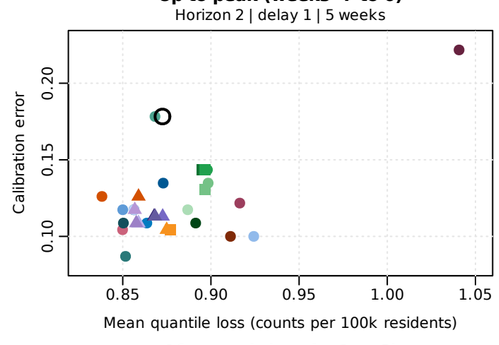


Feedback delay 1

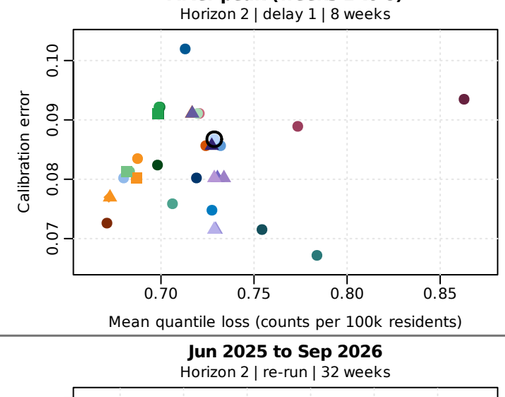
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks

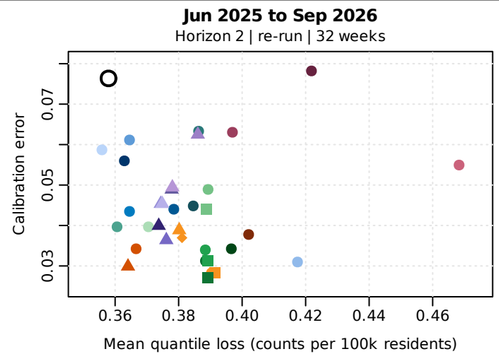


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

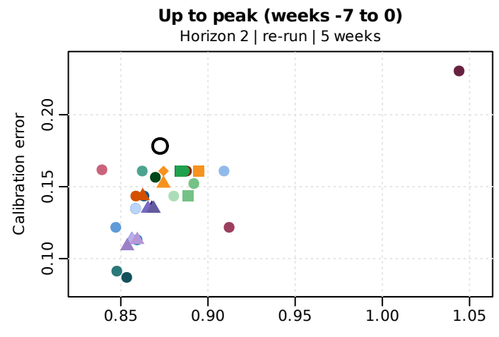


Feedback delay 0 + revisions (re-run)

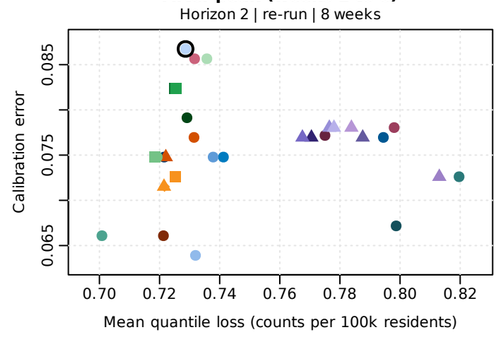
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks



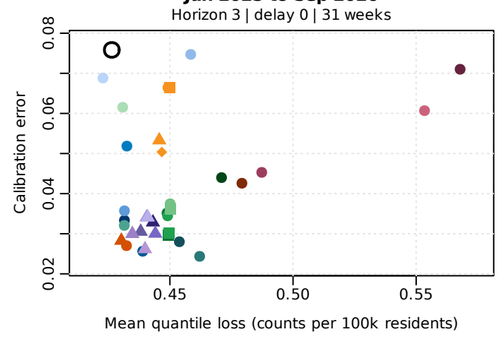
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



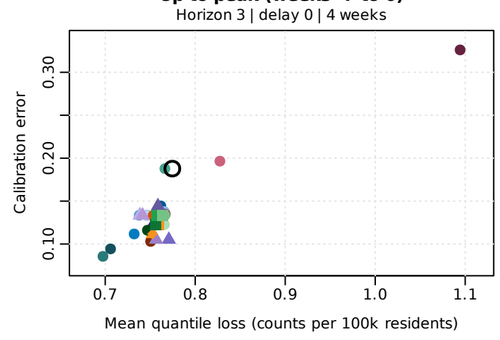
Horizon 3

Feedback delay 0

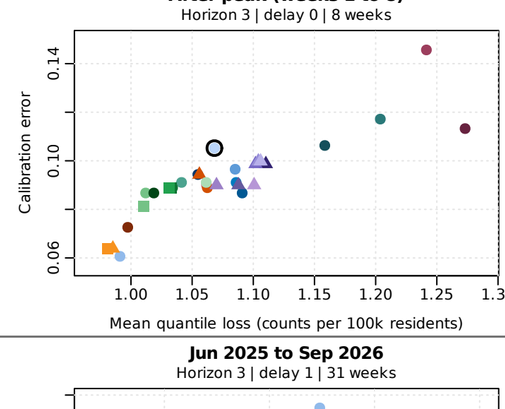
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks

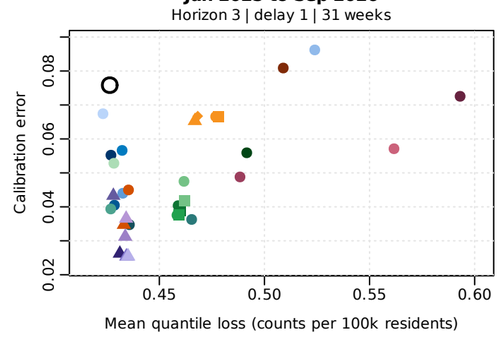


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

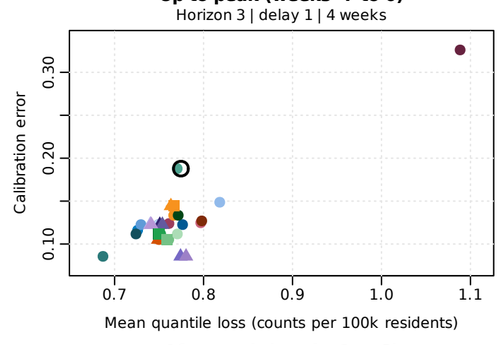


Feedback delay 1

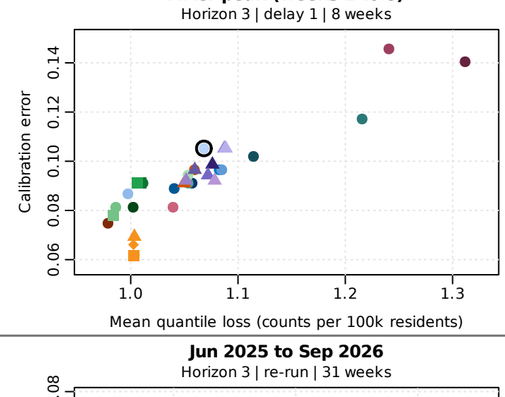
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks

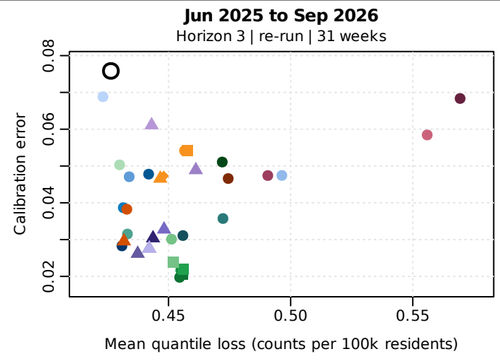


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks

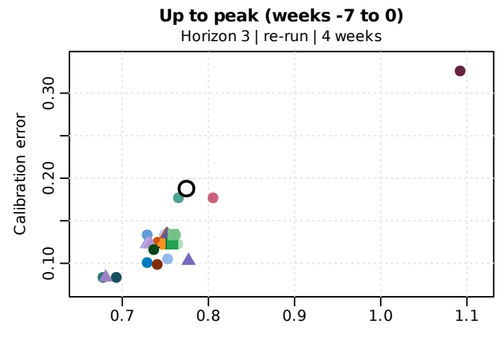


Feedback delay 0 + revisions (re-run)

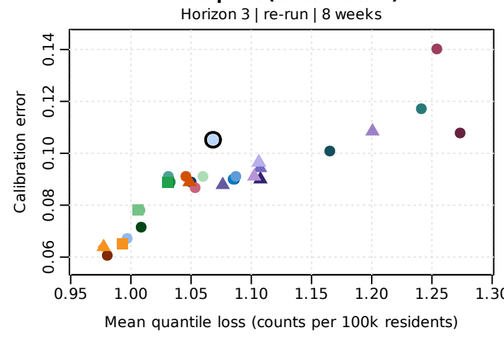
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

slow 0.003, leak 0.1)
)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0.1, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, target signal (C, delta=0.1, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

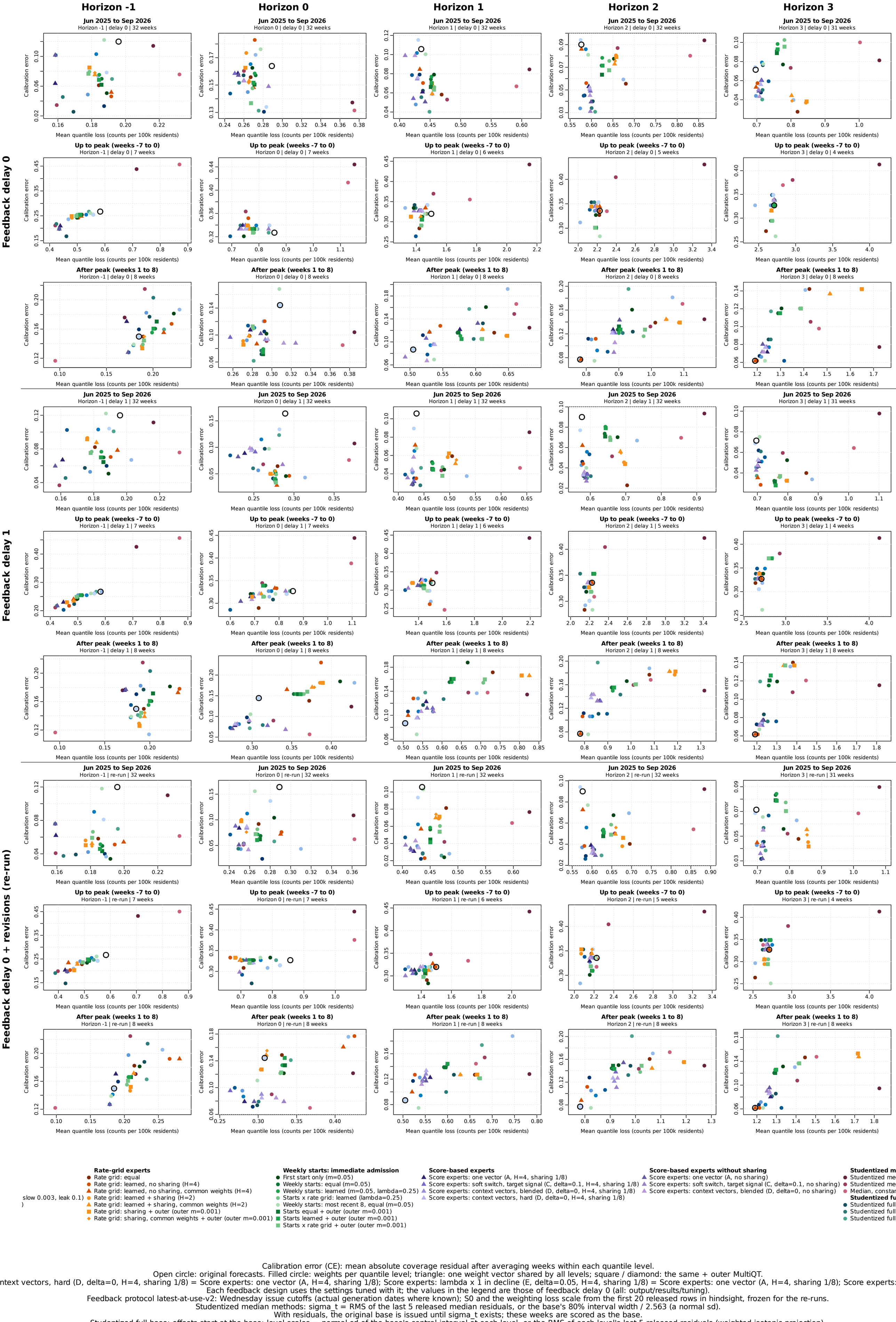
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Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection)

flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo



slow 0.003, leak 0.1)
)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
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Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
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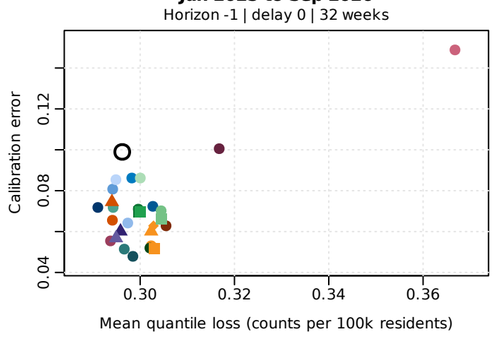
flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

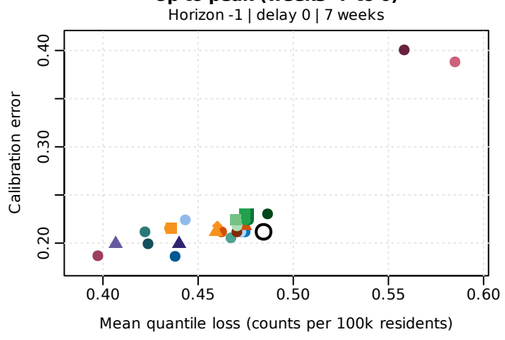
Horizon -1

Feedback delay 0

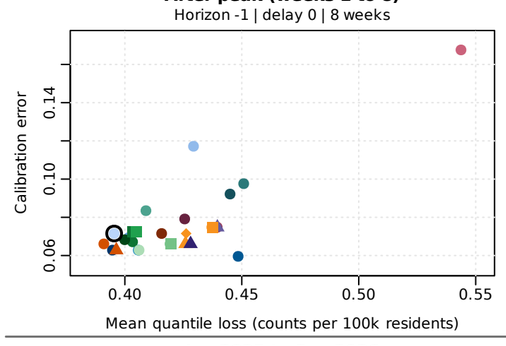
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks

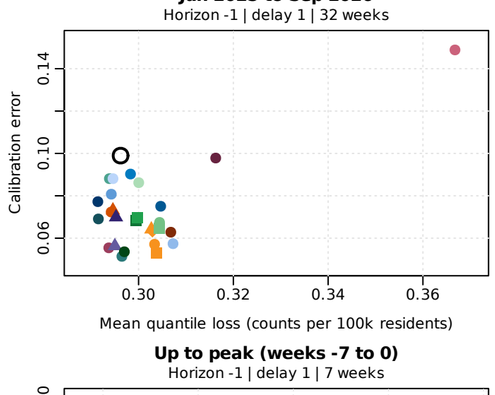


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

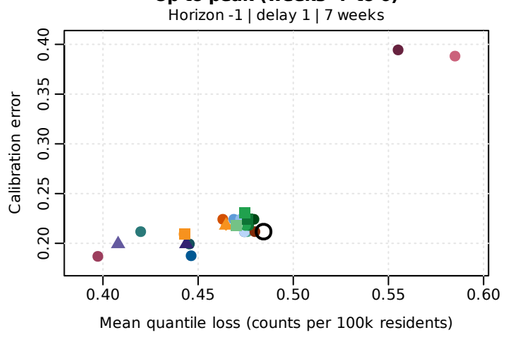


Feedback delay 1

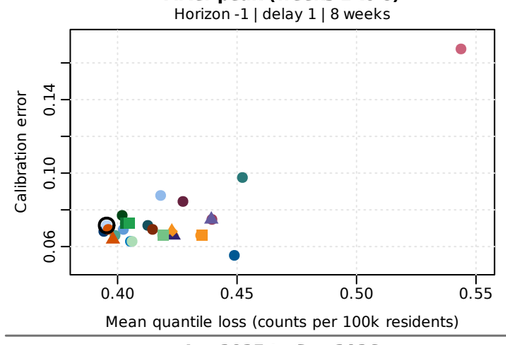
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks

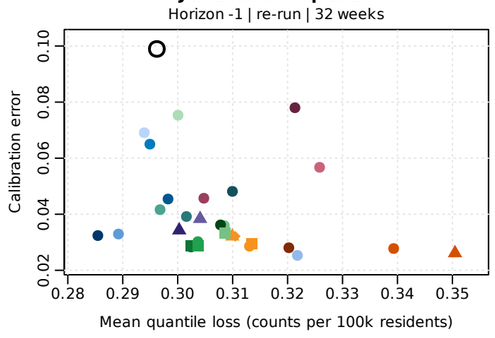


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

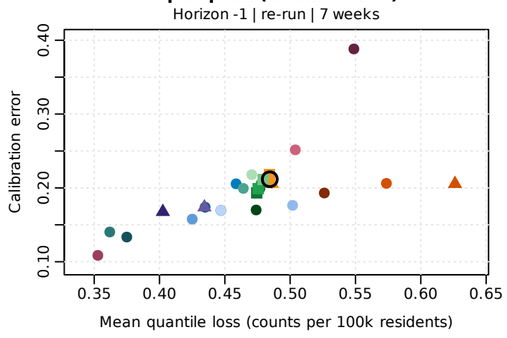


Feedback delay 0 + revisions (re-run)

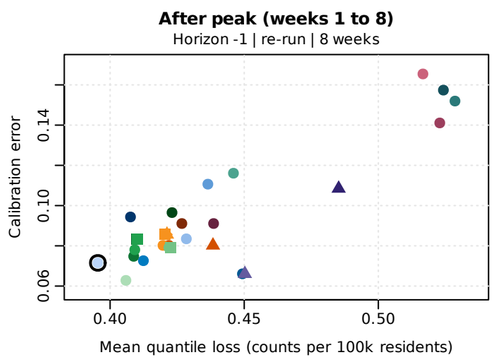
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks



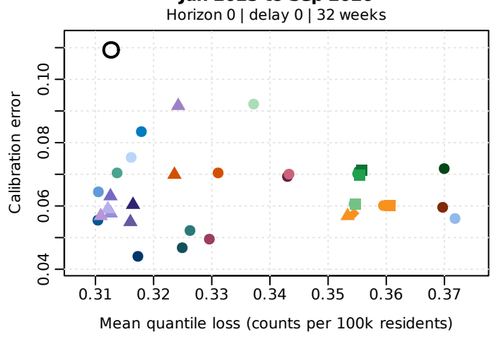
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



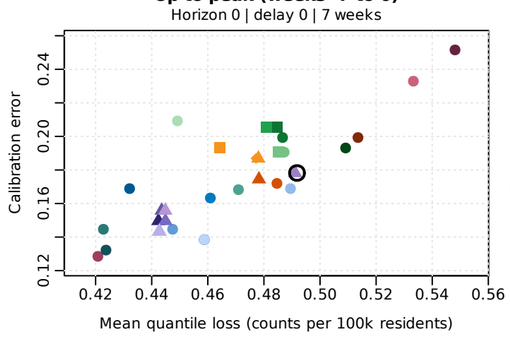
Horizon 0

Feedback delay 0

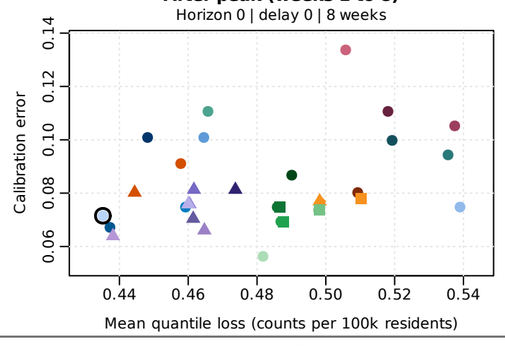
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks

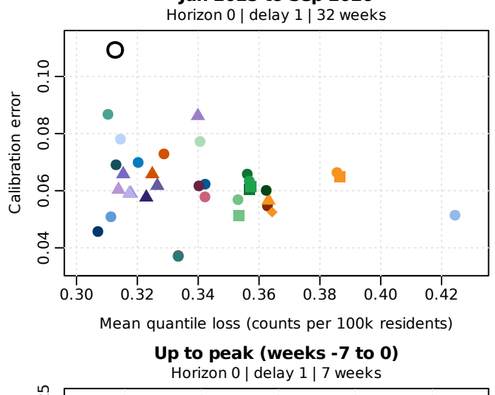


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

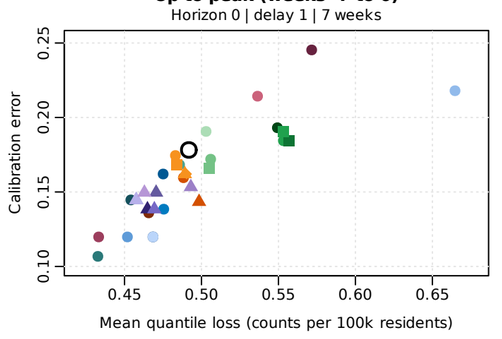


Feedback delay 1

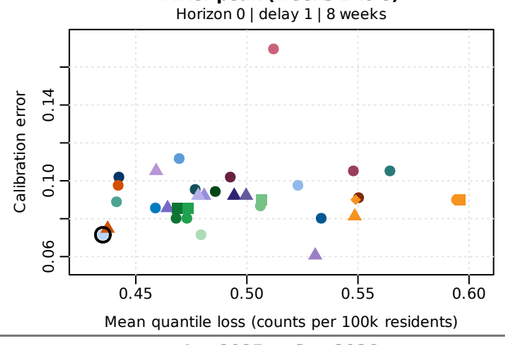
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks

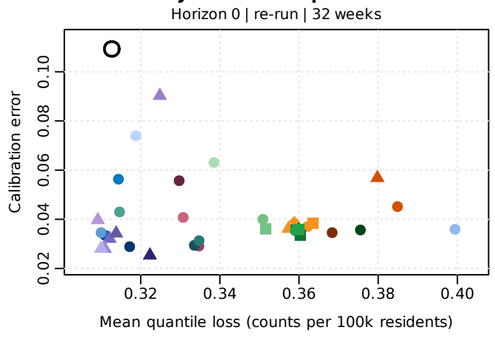


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

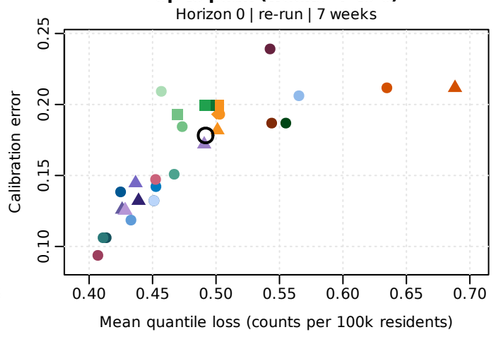


Feedback delay 0 + revisions (re-run)

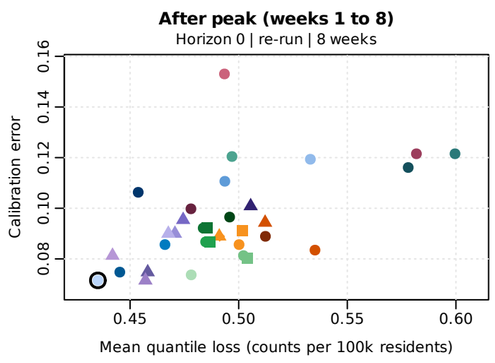
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks



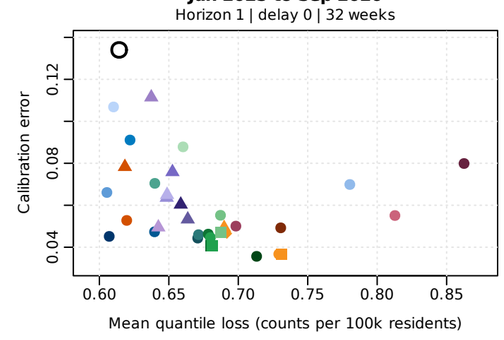
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



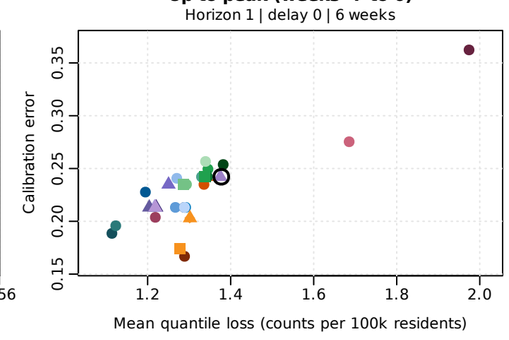
Horizon 1

Feedback delay 0

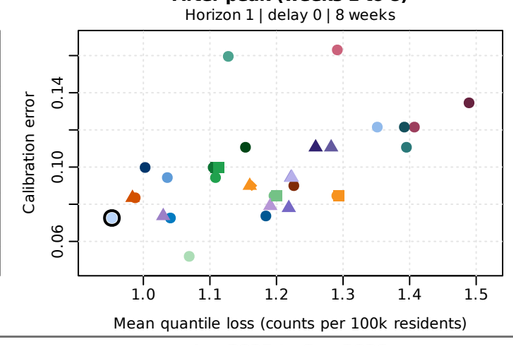
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks

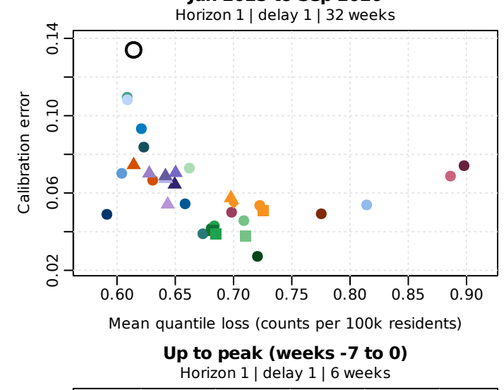


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

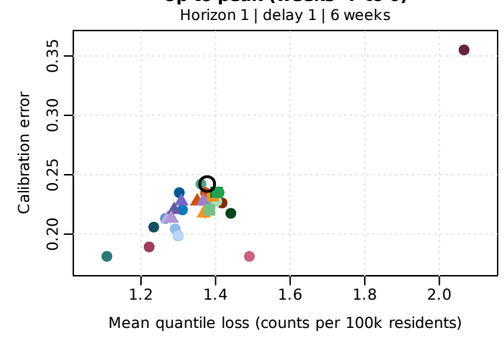


Feedback delay 1

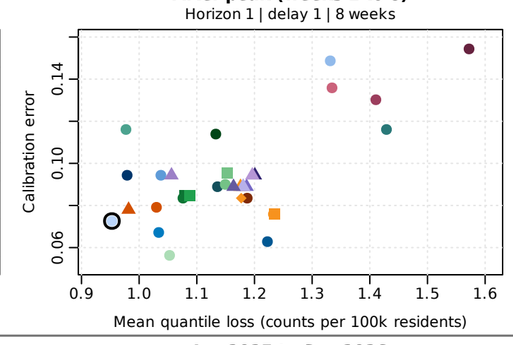
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks

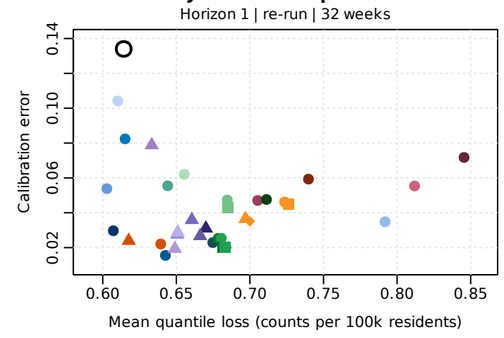


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

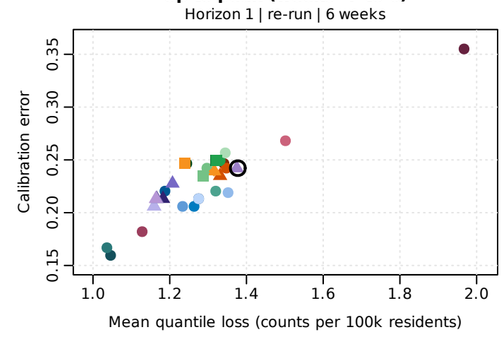


Feedback delay 0 + revisions (re-run)

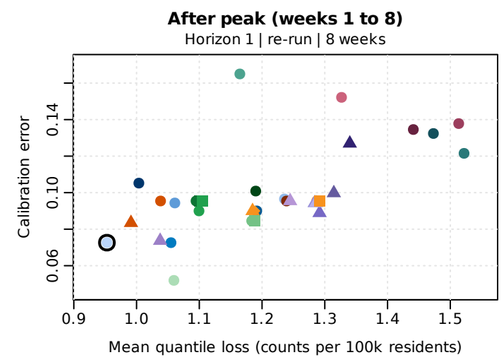
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks



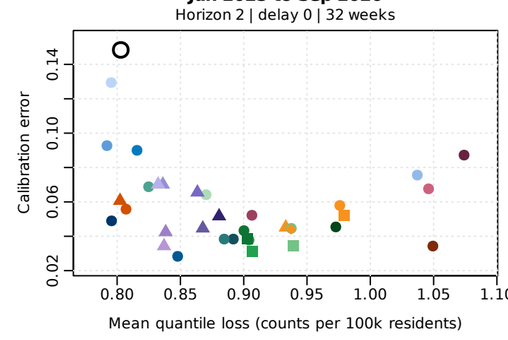
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



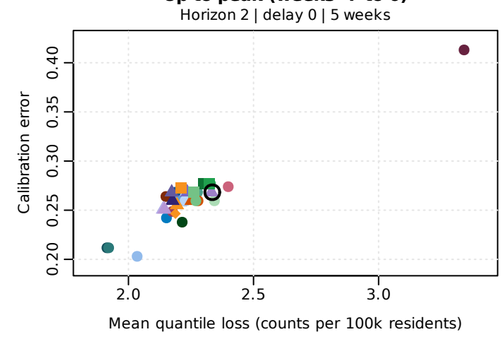
Horizon 2

Feedback delay 0

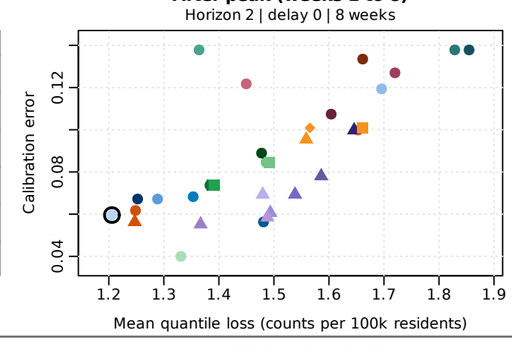
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks

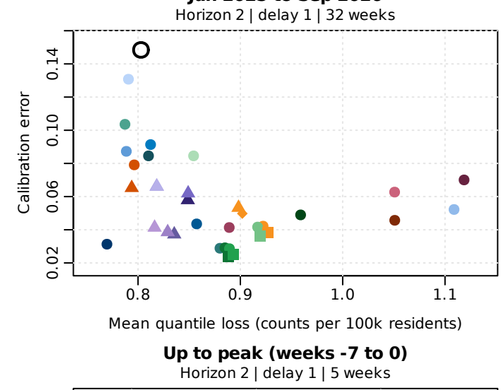


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

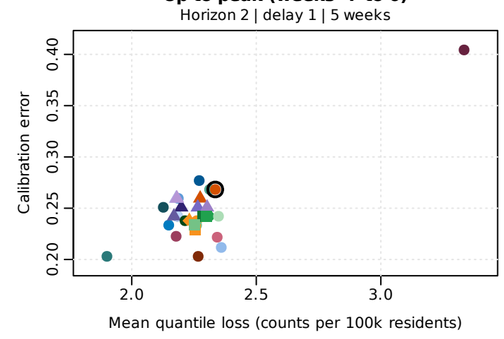


Feedback delay 1

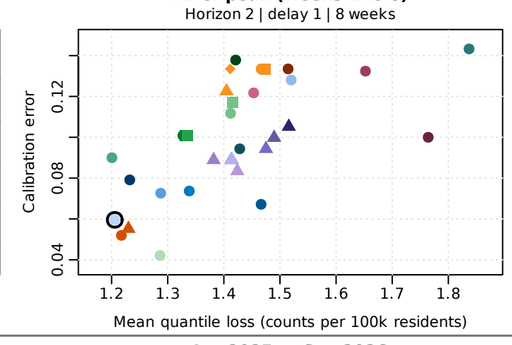
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks

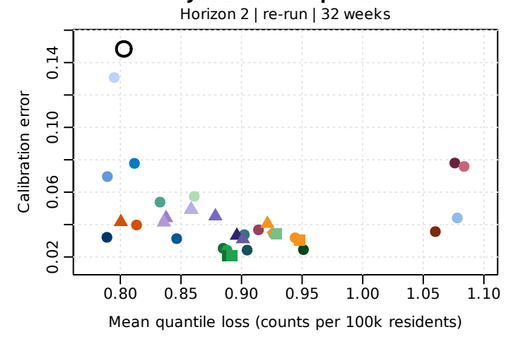


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

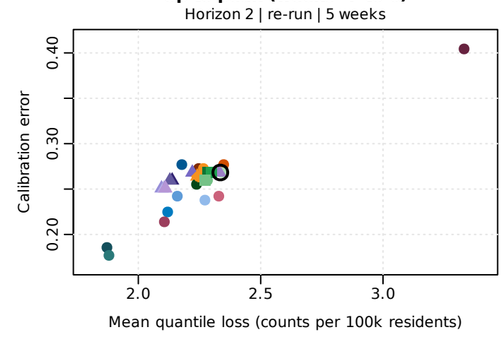


Feedback delay 0 + revisions (re-run)

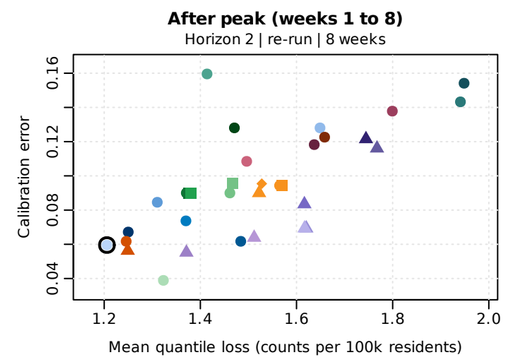
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks



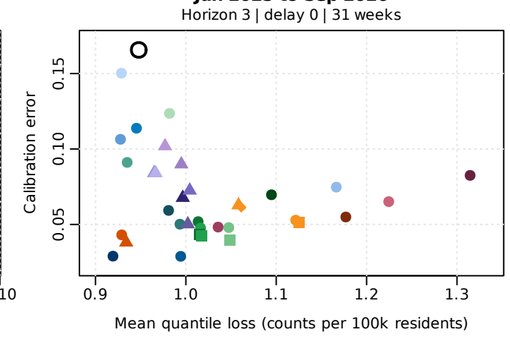
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



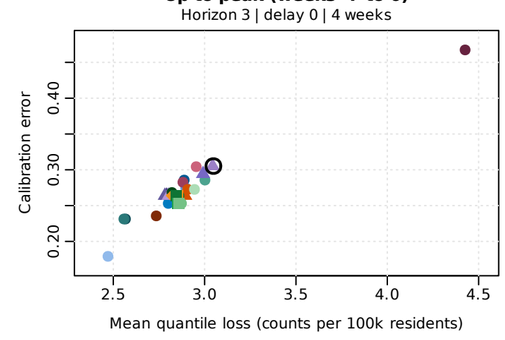
Horizon 3

Feedback delay 0

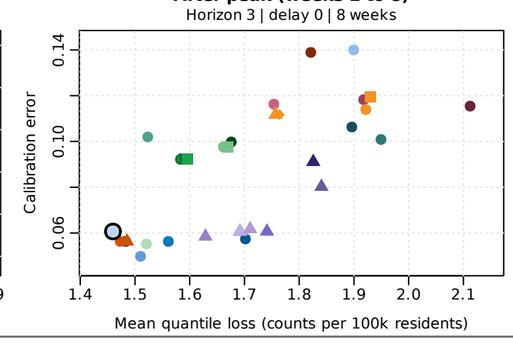
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks

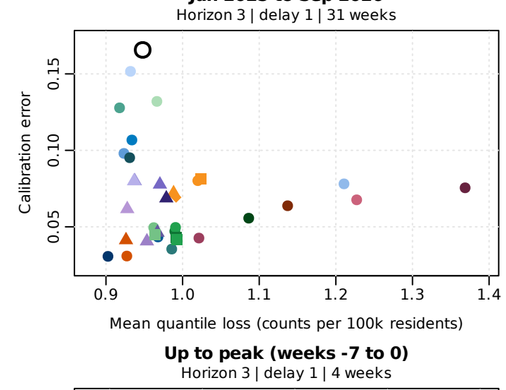


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

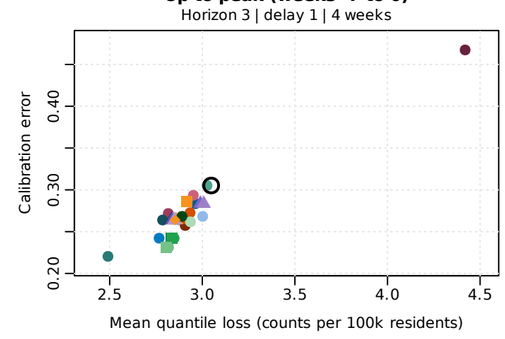


Feedback delay 1

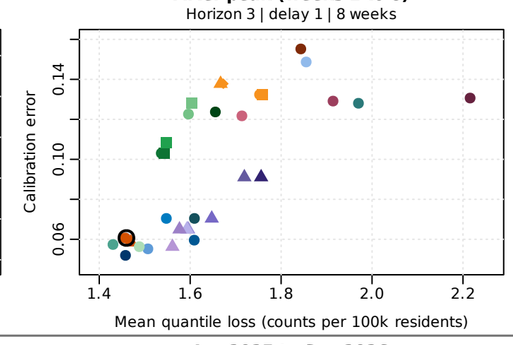
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks

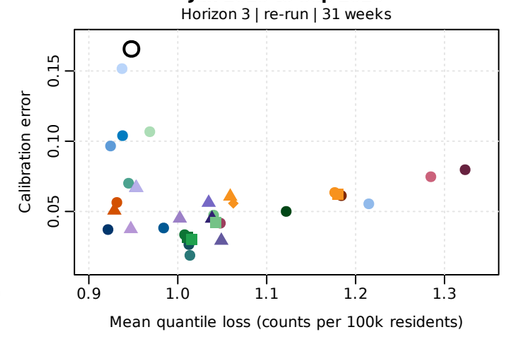


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks

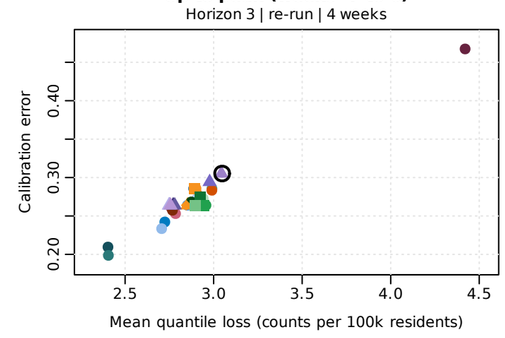


Feedback delay 0 + revisions (re-run)

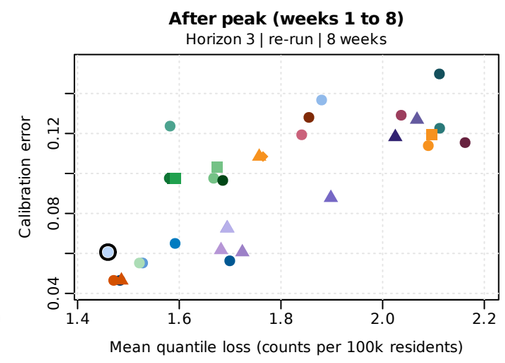
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks



Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

slow 0.003, leak 0.1)
)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Weekly starts: most recent 8, equal (m=0.05)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Score-based experts
- Score experts: one vector (A, H=4, sharing 1/8)
- Score experts: soft switch, target signal (C, delta=0.1, H=4, sharing 1/8)
- Score experts: context vectors, blended (D, delta=0, H=4, sharing 1/8)
- Score experts: context vectors, hard (D, delta=0, H=4, sharing 1/8)

Score-based experts without sharing
- Score experts: one vector (A, no sharing)
- Score experts: soft switch, target signal (C, delta=0.1, no sharing)
- Score experts: context vectors, blended (D, delta=0, no sharing)

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Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
ntext vectors, hard (D, delta=0, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts: lambda x 1 in decline (E, delta=0.05, H=4, sharing 1/8) = Score experts: one vector (A, H=4, sharing 1/8); Score experts:
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection)